

CASCO Claims-Cost & Underwriting Pricing Model

TBC Insurance — Technical Task · End-to-End Project Presentation

Domain: Motor (CASCO) insurance · **Two parts:** A — Segment rating · B — Advanced GLMs/GBM **Tests:** 48 tests · 11 modules · 0 failures · **Languages:** English + Georgian (KA)

Agenda

- 1. Business problem & objectives
- 2. Data sources & schema
- 3. Solution architecture (pipeline)
- 4. Part A — Segment experience-rating
- 5. Part B — Advanced statistical models
- 6. Key modelling decisions & "why"
- 7. Engineering strength: tests, EDA, experiments
- 8. What we tested (full matrix)
- 9. Results, findings & insights
- 10. Conclusion & next steps

1 · Business Problem & Objectives

CASCO is comprehensive motor insurance. The book is **under-priced**: portfolio loss ratio is **70.6%** against a **60%** management target — roughly an **18% average under-pricing**.

Loss Ratio (LR) = Incurred Claims ÷ Earned Premium. **Pure premium** estimates $E[\text{cost}] = E[N] \cdot E[S]$ where N = claim count (frequency) and S = claim amount (severity). **Technical premium** = $\text{model_pp} / \text{target_LR}$.

Objective	What we deliver
Diagnose	Segment the book on 4 rating dimensions; surface hot/cold segments driving 70.6%.
Indicate	Credibility-weighted, capped (±50%) rate corrections vs. target loss ratio.
Model	Individual-risk frequency × severity models, validated out-of-time , with rate indication + elasticity.

2 · Data Sources & Schema

All raw inputs live in `data/` ; a single trusted cleaning/join layer (`src/load_data.py`) produces one analysis-ready policy-level frame.

File	Role	Key content
Policies_1805_v4.csv	Policy master	premium, sum insured, vehicle, driver age, effective date, deductible
Claims_1805_v2.csv	Claims	claim amounts, dates, linked policy
Clients_1805_v3.csv	Client master	client type (physical/legal), personal-use flag
enrcols.csv	Enrichment	deductible category, eligible driver count, min driver age, car age
franchise_field_categories.csv	Lookup	deductible category mapping
fx_rates.csv	FX	currency → GEL (USD sum-insured → GEL)

Cleaning rules: keep CASCO + physical + personal-use; validate claims; convert sum-insured to GEL; compute `earned_premium`; aggregate claims per policy (zeros filled); compute `loss_ratio` only where `earned_premium > 0` (safe divide → NaN); derive `exposure = earned_premium.clip(lower=0)` (GLM offset / GBM weight); enrich with `deductible_type`, `eligible_driver_count`, `min_driver_age`, `carage`.

This single layer is the only place raw data is touched — every downstream module consumes the same clean frame, so changes cascade predictably.

3 · Solution Architecture

Two cooperating parts share the cleaned frame. Part A = fast, explainable segment view; Part B = individual-risk models validated out-of-time.

Part A — Segment experience-rating (`run.py` orchestrates): `load_data` → `segments` → `metrics` → `recommend` → `run` (CSV + Excel)

Part B — Advanced individual-risk models: `features` → `frequency` → `severity` → `pure_premium` → `credibility` → `pricing` → `validation`

- **Time-based split:** train `efdate < 2025-01-01`, test `efdate ≥ 2025-01-01` (avoids leakage).
- **Exposure as offset:** `offset = log(exposure)` where `exposure = earned_premium`; re-applied at predict.
- **Outputs:** 1,264 segments, portfolio summary, 6 breakdown sheets, model metrics & monitoring tables.

4 · Part A — Segment Experience-Rating

Segmentation (`segments.py`) — four rating dimensions, collapsed for stability: - **Age:** 18–20 · 21–25 · 26–29 · 30–40 · >40 - **Vehicle type:** SUV-type · Sedan · Coupé/Cabrio (collapsed to 3) - **Deductible:** Yes / No - **Sum insured:** USD bands from the brief (≤5k · 1k-wide bands 5k–50k · >50k)

→ **1,264** composed segments.

Metrics (`metrics.py`): `aggregate_by` computes per-segment LR/rate/frequency/severity; `safe_div` returns NaN (not inf) on zero exposure; `aggregate_overall` gives portfolio totals.

Recommendations (recommend.py): correction = credibility-weighted experience vs. target_loss_ratio = 0.60 ; capped ±50%; thin segments shrink toward the mean via credibility z ; action label (increase/decrease/hold) attached.

All corrections are point estimates that ignore elasticity — a documented limitation.

5 · Part B — Advanced Statistical Models

- **Frequency (frequency.py):** Poisson & NegBin GLM with offset = log(exposure) ; LightGBM Poisson GBM using log_exposure as a feature; NegBin started from Poisson MLE with alpha=0.1 (never 0 → avoids log(0) BFGS blow-up); **offset re-applied at predict**.
- **Severity (severity.py):** models amount per claim; LogNormal GLM + LightGBM Gamma GBM; **Gamma GLM dropped** (IRLS/BFGS diverges on heavy tail).
- **Pure premium (pure_premium.py):** $E[\text{cost}] = E[N] \cdot E[S]$ — multiplicative Poisson×LogNormal **and** direct Tweedie (power 1.5) on incurred_claim ; calibration_factor = mean(actual \$)/mean(predicted \$) .
- **Credibility (credibility.py):** Bühlmann-Straub $z_i = n_i / (n_i + k)$, $k = EPV/VPP$; n_i = **policy count** (not dollar exposure); shrinks observed PP toward portfolio mean.
- **Pricing (pricing.py):** technical premium = model_pp / target_LR (0.65 in Part B); rate change capped ±50%; elasticity-based volume response (−0.5); calibration factor applied pre-indication.
- **Validation (validation.py):** monthly actual-vs-predicted LR monitoring on hold-out; out-of-time insurance Gini; level-calibration check (ratio ≈ 1.0).

6 · Key Modelling Decisions & "Why"

Decision	Why
Time-based split (not random)	Random split leaks future policies into training and inflates Gini.
offset = log(earned_premium)	Standardises to one unit of exposure; re-applied at predict or model predicts rate×exposure, not expected count.
Drop Gamma GLM for severity	IRLS/BFGS diverges on heavy tail; LogNormal GLM + Gamma GBM are stable.
Multiplicative freq×sev and Tweedie	Decomposes cost driver + cross-check; Tweedie models zero-inflated claims natively.
Credibility uses policy count n	Dollar exposure would make every segment "fully credible"; count reflects real volume.
±50% cap + credibility	Prevents thin segments from destabilising extreme corrections.
Calibration factor before indication	Multiplicative model under-predicts average PP (factor ≈ 1.68); corrects level, not shape.
Two target LRs (0.60 / 0.65)	Separate management parameters; both set before deployment.

7 · Engineering Strength — Tests, EDA, Experiments

A model you cannot repeat or defend is not shippable.

- **Tests (tests/):** **48 tests · 11 modules · 0 failures.** Each file documents *why* it exists; the suite already caught a NegBin `alpha=0` divergence (20-min lock-up) and an experiments `KeyError: 'segment'`.
- **EDA (src/eda.py):** 9-step function (overview, frequency, severity, exposure, segments, time trend, feature relationships, zero-claim deep-dive, credibility preview) — each *shows* the facts.
- **Experiments (src/experiments.py):** 8 what-if probes that *prove* each choice: Gamma GLM divergence (EXP 2), `log_exposure` ablation (EXP 5), policy-count vs dollar credibility (EXP 6), calibration flipping the indication sign (EXP 7), NegBin vs Poisson (EXP 8).

Process rule: any change in `load_data.py`, a model, or a metric is "ready" only when `python -m tests.run_all` is green and `python -m src.experiments` still completes.

8 · What We Tested — Full Matrix

Module	# Tests	Asserts
test_load_data	8	policy-level & non-empty; CASCO-only; physical; personal-use; positive finite SI; claims aggregated w/ zeros; deductible yes/no; LR only where earned>0
test_segments	5	4 dims present; age bins gap-free; franchise binary; vehicle collapsed to 3+; key unique per combo
test_metrics	4	safe_div → NaN not inf; core metrics present; internally consistent; known portfolio LR
test_recommend	5	corrections within cap; credibility in [0,1]; shrinks thin segments; hot segment +correction; action label consistent
test_features	5	time split no overlap & respects cutoff; y_sev only where claim; feature engineering; matrix shapes; exposure non-negative
test_frequency	4	Poisson deviance ≥ 0 ; offset re-applied at predict; train/evaluate runs & discriminates
test_severity	4	Gini of constant = 0; Gamma deviance ≥ 0 ; training target no zeros; train/evaluate runs & discriminates
test_pure_premium	4	Tweedie deviance ≥ 0 ; calibration factor sensible; predict aligned to exposed rows; discriminates
test_credibility	3	Z in [0,1] & $k>0$; mean Z shows real shrinkage; blended Gini beats raw out-of-time
test_pricing	3	rate change capped; volume = elasticity × rate change; calibrated indication positive on avg
test_validation	3	monitoring report shape & columns; holdout Gini > 0; level calibration ≈ 1
Total	48	11 modules · 0 failures

9 · Results, Findings & Insights

Portfolio (Part A): LR **70.6%** (vs 60% target) · **119,745** policies · Earned premium **GEL 172.4M** · Incurred **GEL 121.7M**.

Model discrimination (out-of-time, Part B):

Model	Gini	Note
Frequency — Poisson GLM	0.3909	baseline GLM
Frequency — LightGBM (Poisson)	0.4085	beats GLMs
Severity — LogNormal GLM	0.2664	interpretable base
Pure premium — Freq×Sev (GLM)	0.4086	strongest ranker
Pure premium — Tweedie GBM	0.3555	competitive on deviance
Credibility — blended	0.0805	vs raw 0.0678 (improvement)

Key insights: - **Young drivers (21–29)** worst on LR *and* already pay highest per-policy rate ($\sim 2\times >40$) — book still loses money on them. - **No-deductible policies** ~ 19 pts worse (0.743 vs 0.555) at the *same* average rate — deductible not priced. - **Low sum-insured cars** most under-priced; rate curve is inverse to risk. - **Test calibration:** actual LR 0.8225 vs predicted 0.7838 (target 0.65) → well-calibrated in *level* (≈ 1.05); book $\sim 17\text{--}22\%$ above target. - Avg indicated rate change **+18.5%**; volume **−9.3%**; 22,505 policies ↑, 5,735 ↓.

10 · Conclusion & Next Steps

Delivered: explainable segment rating (1,264 segments) + model-based individual-risk pricing; out-of-time validated with full test/EDA/experiment harness; complete Georgian translation reviewed and verified (0 test failures).

Next steps: 1. Set management target loss ratios (cost + profit load) before deployment. 2. Calibrate full-credibility standard to portfolio volatility. 3. Validate corrections on a hold-out; simulate premium/LR response. 4. Deploy `validation.py` monitoring dashboard; back-test indication vs realised LR. 5. Consider GLMM / boosted hybrid and a proper Tweedie power search.

Bottom line: the book is under-priced by $\sim 18\%$; the model explains *where* (young drivers, no-deductible, low sum-insured) and indicates risk-adequate, credibility-capped corrections — reproducible, tested, and documented in both English and Georgian.

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